

Joris Gentinetta

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Work Experience

Software Engineer in Image Analysis, Roche Institute of Human Biology

06/25-now

Goal: Establishing state-of-the-art image analysis capabilities, screen analysis and workflow manager (Fractal).

Outcome: Worked on 15+ projects | Established capabilities: graph-based cell-to-organoid construction, particle/cell/organoid tracking, cell typing, membrane structure and developmental patterning estimation | Feature extraction and statistical analysis for compound screens | Interactive curation tooling | Led the Fractal dev team for three months (3 people), from initial experiments to the first cloud-based workflow | Supervised analytics and software engineering interns on image viewer development and foundation model projects. | Co-authored manuscript under review at *Cell*.

Internship, Roche Institute of Human Biology

10/24-05/25

Goal: Cell tracking in 3D microscopy data.

Outcome: Established cell tracking in 3D | Productionalized preprocessing for a custom oblique plane microscope

Research Assistant, ETH/UZH Institute of Neuroinformatics

09/22-02/24

Goal: Building a robot to interact with Marmoset monkeys to study the brain mechanisms for imitation learning.

Outcome: Initiated and led the project | Developed a fully tendon driven 16 DOF robot | Supervised a bachelor's thesis and an intern.

Internship, IBM Research Zurich – Awarded Best Intern 2023

11/22-10/23

Goal: Improving current learning algorithms using insights from neuroscience and in-memory computing.

Outcome: Created a PyTorch framework for backpropagation-in-time training on both simulated and real in-memory computing hardware | Reduced the memory complexity of the OSTL learning algorithm for RNNs from quadratic to linear | Built a Ray interface for the LSF cluster | [Published at AI/CAS 2023](#), [Acknowledged in Nat. Commun.](#)

Internship, ABB Semiconductors / Hitachi Energy

09/21-02/22

Goal: Defect classification and fault-pattern detection on semiconductor chips.

Outcome: Deployed a productive defect-analysis pipeline with automated training, human-in-the-loop relabeling of uncertain samples, spatial fault-pattern detection, and GUI-based result inspection.

Education

MSc Data Science, ETH / MIT / SNU

09/22-09/24

Neural Information Processing track: Focus on artificial intelligence and brain-like computation.

Master's thesis at MIT in spring 2024 | Exchange at SNU, South Korea in spring 2023

BSc Electrical Engineering and Information Technology, ETH

09/18-09/22

Projects in data analysis/machine learning as well as neuroinformatics.

Biology and Chemistry Matura, Stiftsschule Einsiedeln

09/11-09/17

Finalist in the Swiss Biology Olympiad 2017

Selected Projects

EMG Based Prosthesis Control [Master's Thesis, MIT Biomechatronics Group]

2024

Goal: Improving neuromuscular modeling-based prosthesis control with neural network-based EMG disentanglement and a novel data acquisition scheme based on pose estimation.

Outcome: Developed an active learning system for controller retraining | Established pose estimation for training data generation | Reduced training time from a full day to under an hour | Mentored 3 undergraduate students | [First-author publication in IEEE TNSRE](#)

Optogenetic Stimulation [Group project, ETH Laboratory of Biosensors and Bioelectronics]

2020

Goal: Developing an LCD based and a projector based optogenetic setup for in vitro stimulation of neurons.

Outcome: Initiated the project and coordinated 5 people | Delivered a functional single-cell-resolution optogenetic stimulation system | Research group transitioned to optogenetics